

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-6682
Project Name	COMIC CRAFT AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited concurrent-user capacity	Response times may increase when many users create or view comics at once.	High
2	Growing storage needs	Comic images, generated panels, and project files can consume substantial storage.	Medium
3	Dependence on AI service availability	Generation may be delayed or unavailable if the external AI service is slow or down.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	A single application instance may be sufficient for initial usage.	Deploy multiple app instances behind a load balancer and add capacity as demand grows.
2	Data Storage	Project data and generated assets use the application's configured storage.	Use managed database storage and object storage for generated images, with backups and retention rules.
3	Performance	AI generation and image processing can take time; requests may compete for resources.	Use background jobs/queues for generation, caching for repeated reads, and monitor latency and errors.
4	Security	Application-level validation and access controls are needed.	Add strong authentication and authorization, input validation, secret management, rate limits, HTTPS, and regular dependency updates.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Improve reliability, validation, and user feedback during comic generation.	1–2 months	Fewer failed requests and a smoother creation workflow.
Phase 3	Add scalable background generation, asset storage, and performance monitoring.	3–4 months	Better handling of larger projects and more simultaneous users.
Phase 4	Expand collaboration and project management features; strengthen security and backups.	5–6 months	A more secure, maintainable platform ready for broader use.